

Saurabh Kumar

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WORK EXPERIENCE

- Research Intern – Software Engineering & QA Diagnostics | SRM University-AP** Jun 2025 – Jul 2025
- Engineered high-performance **Python, Java, and C++** software pipelines leveraging OOP design, advanced data structures, and tensor optimizations to achieve **1.84 sec/inference throughput**.
 - Designed automated **unit testing, integration testing, and defect logging harnesses** to validate software reliability, memory efficiency, and execution speed across production builds.

- Core Team Member | Google Developer Groups (GDG), SRM-AP** Aug 2025 – Present
- Mentored **150+ developers** across **5+ technical workshops** covering **Java, Python, REST APIs, Microservices Architecture, Postman, and Git** workflows — driving a **30% community growth**.
 - Facilitated developer hackathons, managing technical QA logistics, participant tracking metrics, and cross-functional team deliverables under tight milestone deadlines.

PROJECTS

- Shiksha Hub – Microservices Architecture & Cloud AI Platform** [Live](#) | [GitHub](#)
Java, Python, REST APIs, Microservices, AWS (EC2, S3, Lambda), Docker, Kubernetes, PostgreSQL, MongoDB, Redis, Git
- Architected asynchronous **microservices communicating via REST APIs** deployed on **AWS Cloud (EC2, S3, Lambda)**, handling high-throughput traffic with **sub-100ms response times**.
 - Containerized multi-service applications using **Docker & Kubernetes**, establishing scalable deployments with **Redis caching** and multi-database architecture (**PostgreSQL, MongoDB**).
 - Applied advanced **Data Structures and Algorithms (DSA)** for queue management, implementing RBAC security, structured logging, Postman API test suites, and CI/CD quality guardrails via **GitHub Actions**.

- RepoLens AI – Graph RAG Code Intelligence & Automated Bug Detection Platform** [GitHub](#)
Python, FastAPI, React.js, Neo4j, ChromaDB, Docker, Pytest, Postman, REST APIs, JIRA-style Defect Logging, Git
- Architected a multi-service code intelligence system incorporating **7 AI agentic tools** for automated **bug-pattern detection, request-flow tracing, and structural dependency verification**.
 - Built automated **API contract tests, input validation guardrails, and stress testing harnesses** using Pytest & Postman to validate backend endpoint reliability under high concurrency.
 - Formulated concise, formatted bug summary reports with execution tracebacks, severity ratings, and reproduction steps to ensure zero regression across multi-language codebases.

- Email Classifier – Scalable Microservice & REST API System** [GitHub](#)
Python, Java, REST APIs, Microservices, FastAPI, Scikit-learn, Docker, Pytest, Pydantic, Git
- Built a modular backend microservice in **Python & REST APIs**, containerizing the deployment pipeline using **Docker** for seamless CI/CD integration.
 - Engineered data processing and model evaluation pipelines, handling dataset scale of **500k+ records** with **93.97% classification accuracy**.
 - Integrated robust unit testing guardrails with Pytest, Pydantic input validation schemas, and structured exception handling to isolate edge-case anomalies.

EDUCATION

- B.Tech in Artificial Intelligence & Machine Learning – SRM University-AP (CGPA: 8.57 / 10.0)** 2023 – Present
Class 12th Bihar Board [AN College, Patna] 2023
Class 10th CBSE Board [MDJ Public School, Ara, Bihar] 2021

SKILLS

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|----------------------------------|--|
| Languages | Python, Java, C++, C#, JavaScript, SQL, HTML5/CSS3, Shell Scripting |
| Software QA & Testing | Test Case Design, Bug Report Writing (Steps to Reproduce) , Functional Testing, Regression Testing, Exploratory Testing, Boundary & Edge-Case Testing, API Testing (Postman), Defect Lifecycle |
| Gaming & Platforms | Rockstar Titles Knowledge (<i>GTA V, GTA Online, Red Dead Redemption 2</i>), Open-World Mechanics, Mission Scripting Fail-States, Physics Collision Testing, PC (Windows), PlayStation, Xbox, Mobile |
| Tools, Cloud & DBs | Microsoft Office (Excel, Word), JIRA / Defect Databases, Git/GitHub , Postman, PostgreSQL, MongoDB, Redis , Docker, Kubernetes, AWS, PC Diagnostics (GPU drivers, DirectX, Crash Dumps) |
| Core CS & Concepts | Data Structures & Algorithms (DSA) , Object-Oriented Programming (OOP), REST APIs, Microservices, Structured Logging, Technical Documentation |

CERTIFICATIONS

- SAP Certified – SAP Generative AI Developer** [Link](#) Apr 2026
MongoDB Associate Developer [Link](#) Mar 2026
IBM SkillsBuild – Web Development Fundamentals [Link](#) Aug 2025

ACHIEVEMENTS

- 75% Merit Scholarship** at SRM University-AP (CGPA 8.57 / 10.0) for exceptional analytical abilities and academic performance.
- Winner of 6+ Hackathons** developing production-ready microservices, automated diagnostic tools, and software test harnesses.